

AI Report

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AI Generated

AI Probability

100%

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\section{Conclusion} - 6/22/2026

\section{Conclusion}

\label{sec:conclusion}

This paper presented EchoPulse AI, a locally-deployed multimodal music recommender system developed in response to a problem I have observed throughout my 14 years as a performing musician: emerging artists struggle to gain visibility in algorithm-driven recommendation systems, not because their music lacks quality, but because these systems prioritize popularity signals over content affinity.

The architecture integrates a five-stage local pipeline that consolidates heterogeneous music datasets while ensuring reproducibility and traceability without cloud dependencies—a design choice driven by practical constraints of cost and internet instability in our region, but which proved beneficial for research reproducibility. By processing lyrical, acoustic, metadata, and conversational modalities through specialized Transformer encoders fused via cross-attention mechanisms, the system generates unified song representations that prioritize content characteristics over historical popularity.

Empirical validation demonstrated that EchoPulse AI achieves a 45.3\% improvement in Intra-List Diversity and a 157.9\% increase in Emerging Artist Discovery compared to collaborative filtering baselines, while maintaining competitive accuracy metrics (Hit@10 = 0.312). These results exceeded our expectations and confirm that content-aware multimodal fusion can effectively mitigate popularity bias without sacrificing recommendation quality. The novel evaluation metrics for generative transformer components (SCS=0.741, IAR=0.698) further validated that the proposed approach better captures user intent compared to purely interaction-based or LLM-only methods.

However, the current implementation has notable limitations that must be addressed in future work. The system primarily recommends English-language content, requiring adaptation for Spanish-language music and broader multilingual support. The Master Table of 5,710 tracks, while sufficient for validation, represents a carefully curated subset; making the system adaptable to arbitrary music databases requires more flexible ingestion

pipelines. Scaling to industrial-scale catalogs while maintaining multimodal integrity presents significant data engineering challenges, particularly in local environments with limited storage.

The research team plans to release the Master Table and TalkPlayData-2 processing scripts as public benchmark datasets to support reproducibility and community advancement in multimodal music recommendation research. We hope this work contributes to more equitable music recommendation systems that provide emerging artists—whether in Popayán, Colombia, or anywhere else—with fair opportunities to reach audiences who would appreciate their work.

● Sentences that are likely AI-generated.

FAQs

What is GPTZero?

GPTZero is the leading AI detector for checking whether a document was written by a large language model such as ChatGPT. GPTZero detects AI on sentence, paragraph, and document level. Our model was trained on a large, diverse corpus of human-written and AI-generated text with support for English, Spanish, French, German, and other languages. To date, GPTZero has served over 17 million users around the world, and works with over 100 organizations in education, hiring, publishing, legal, and more.

When should I use GPTZero?

Our users have seen the use of AI-generated text proliferate into education, certification, hiring and recruitment, social writing platforms, disinformation, and beyond. We've created GPTZero as a tool to highlight the possible use of AI in writing text. In particular, we focus on classifying AI use in prose. Overall, our classifier is intended to be used to flag situations in which a conversation can be started (for example, between educators and students) to drive further inquiry and spread awareness of the risks of using AI in written work.

Does GPTZero only detect ChatGPT outputs?

No, GPTZero works robustly across a range of AI language models, including but not limited to ChatGPT, GPT-5, GPT-4, GPT-3, Gemini, Claude, and AI services based on those models.

What are the limitations of the classifier?

The nature of AI-generated content is changing constantly. As such, these results should not be used to punish students. We recommend educators to use our behind-the-scenes [Writing Reports](#) as part of a holistic assessment of student work. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Instead, we recommend educators take approaches that give students the opportunity to demonstrate their understanding in a controlled environment and craft assignments that cannot be solved with AI. Our classifier is not trained to identify AI-generated text after it has been heavily modified after generation (although we estimate this is a minority of the uses for AI-generation at the moment). Currently, our classifier can sometimes flag other machine-generated or highly procedural text as AI-generated, and as such, should be used on more descriptive portions of text.

I'm an educator who has found AI-generated text by my students. What do I do?

Firstly, at GPTZero, we don't believe that any AI detector is perfect. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Nonetheless, we recommend that educators can do the following when they get a positive detection: Ask students to demonstrate their understanding in a controlled environment, whether that is through an in-person assessment, or through an editor that can track their edit history (for instance, using our [Writing Reports](#) through Google Docs). Check out our list of [several recommendations](#) on types of assignments that are difficult to solve with AI.

Ask the student if they can produce artifacts of their writing process, whether it is drafts, revision histories, or brainstorming notes. For example, if the editor they used to write the text has an edit history (such as Google Docs), and it was typed out with several edits over a reasonable period of time, it is likely the student work is authentic. You can use GPTZero's Writing Reports to replay the student's writing process, and view signals that indicate the authenticity of the work.

See if there is a history of AI-generated text in the student's work. We recommend looking for a long-term pattern of AI use, as opposed to a single instance, in order to determine whether the student is using AI.